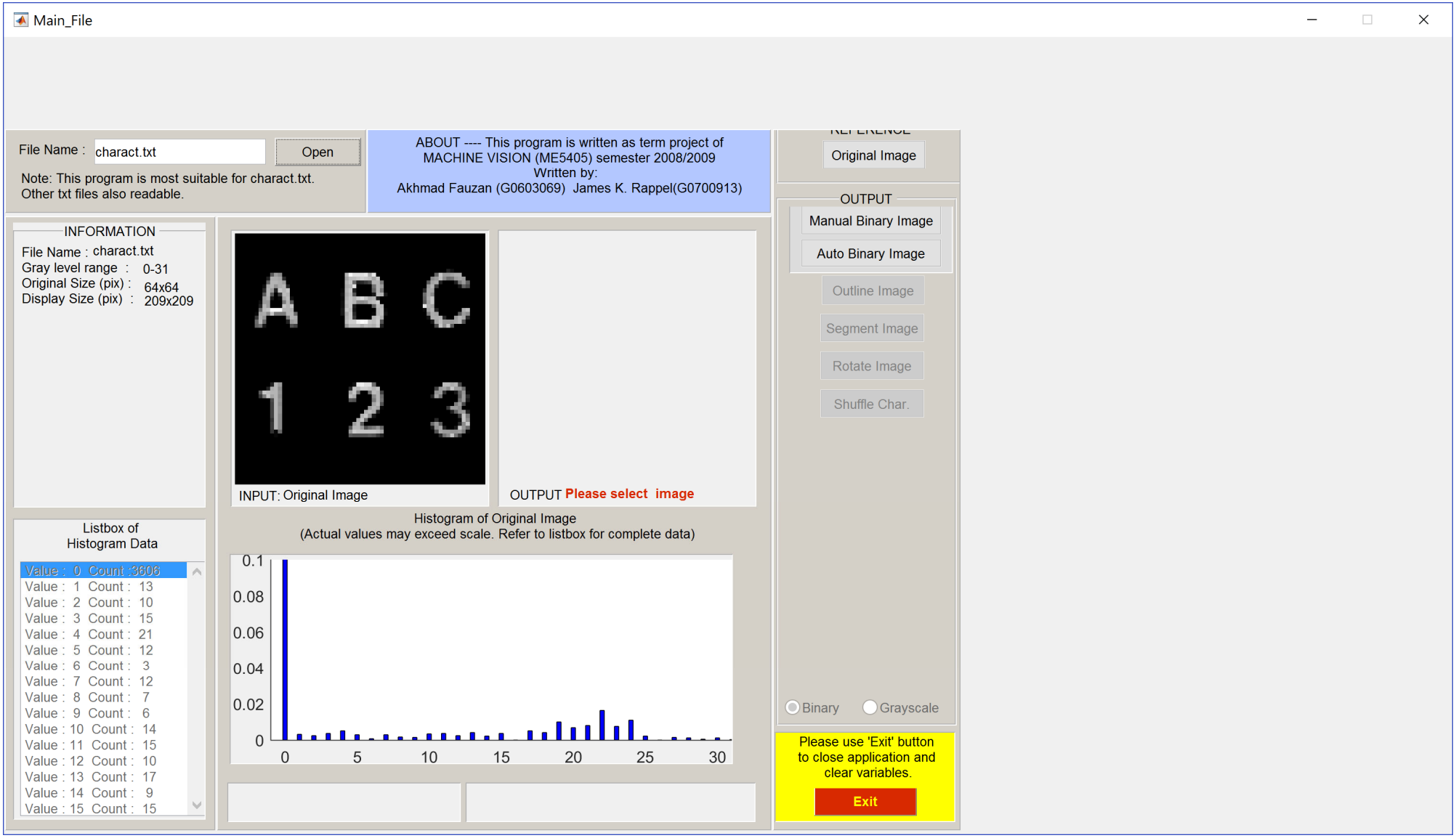
A/Prof CK Chui, NUS ME

Image 1

```
EJNPPRSTSTSTTTURQSVVVVUUUVVVV
```


Main_File
File Name : charact.txt
Open
Note: This program is most suitable for charact.txt.
Other txt files also readable.
ABOUT ---- This program is written as term project of
MACHINE VISION (ME5405) semester 2008/2009
Written by:
Akhmad Fauzan (G0603069) James K. Rappel(G0700913)
Original Image
OUTPUT
Manual Binary Image
Auto Binary Image
Outline Image
Segment Image
Rotate Image
Shuffle Char.
INFORMATION
File Name : charact.txt
Gray level range : 0-31
Original Size (pix) : 64x64
Display Size (pix) : 209x209
A B C
1 2 3
INPUT: Original Image
OUTPUT Please select image
Histogram of Original Image
(Actual values may exceed scale. Refer to listbox for complete data)
0.1
0.08
0.06
0.04
0.02
0
0 5 10 15 20 25 30
Listbox of
Histogram Data
Value : 0 Count :3606
Value : 1 Count : 13
Value : 2 Count : 10
Value : 3 Count : 15
Value : 4 Count : 21
Value : 5 Count : 12
Value : 6 Count : 3
Value : 7 Count : 12
Value : 8 Count : 7
Value : 9 Count : 6
Value : 10 Count : 14
Value : 11 Count : 15
Value : 12 Count : 10
Value : 13 Count : 17
Value : 14 Count : 9
Value : 15 Count : 15
Binary
Grayscale
Please use 'Exit' button
to close application and
clear variables.
Exit

Editor - C:\Users\mpecck\Documents\MATLAB\Athena Jalalian\codes\CH1.m

```
imtool(Iz12)
%=============== translation & scale
M=Mic1;
for k=1:2:5
min=a(2);
for i=1:6
    if M(i)<min
        min=M(i);
        j1=i;
    end
end
M(j1)=a(2);
min=a(2);
for i=1:6
    if M(i)<min
        min=M(i);
        j2=i;
    end
end
M(j2)=a(2);
if Mir1(j1)<Mir1(j2)
    W(k,:)=[Mir1(j1),Mic1(j1),Mar1(j1),Mac1(j1)];
    W(k+1,:)=[Mir1(j2),Mic1(j2),Mar1(j2),Mac1(j2)];
else
    W(k,:)=[Mir1(j2),Mic1(j2),Mar1(j2),Mac1(j2)];
    W(k+1,:)=[Mir1(j1),Mic1(j1),Mar1(j1),Mac1(j1)];
end
end
% W contains the oredered borders of the objects
W(7,:)=[0;0;0;0];
Is1=ones(3*(W(1,3)-W(1,1))-4,6*(W(1,4)-W(1,2))+15,3)*255;
%translation==========
G=0;
for k=1:6
    for i=W(k,1):W(k,3)
        for j=W(k,2):W(k,4)
            if Ib1(i,j)==0
                Is1(i-W(k,1)+W(1,1),j+G,:)=Icol1(i,j,:);
            end
        end
    end
    G=G+W(k,4)+2-W(k+1,2);
end
imtool(Is1)
```

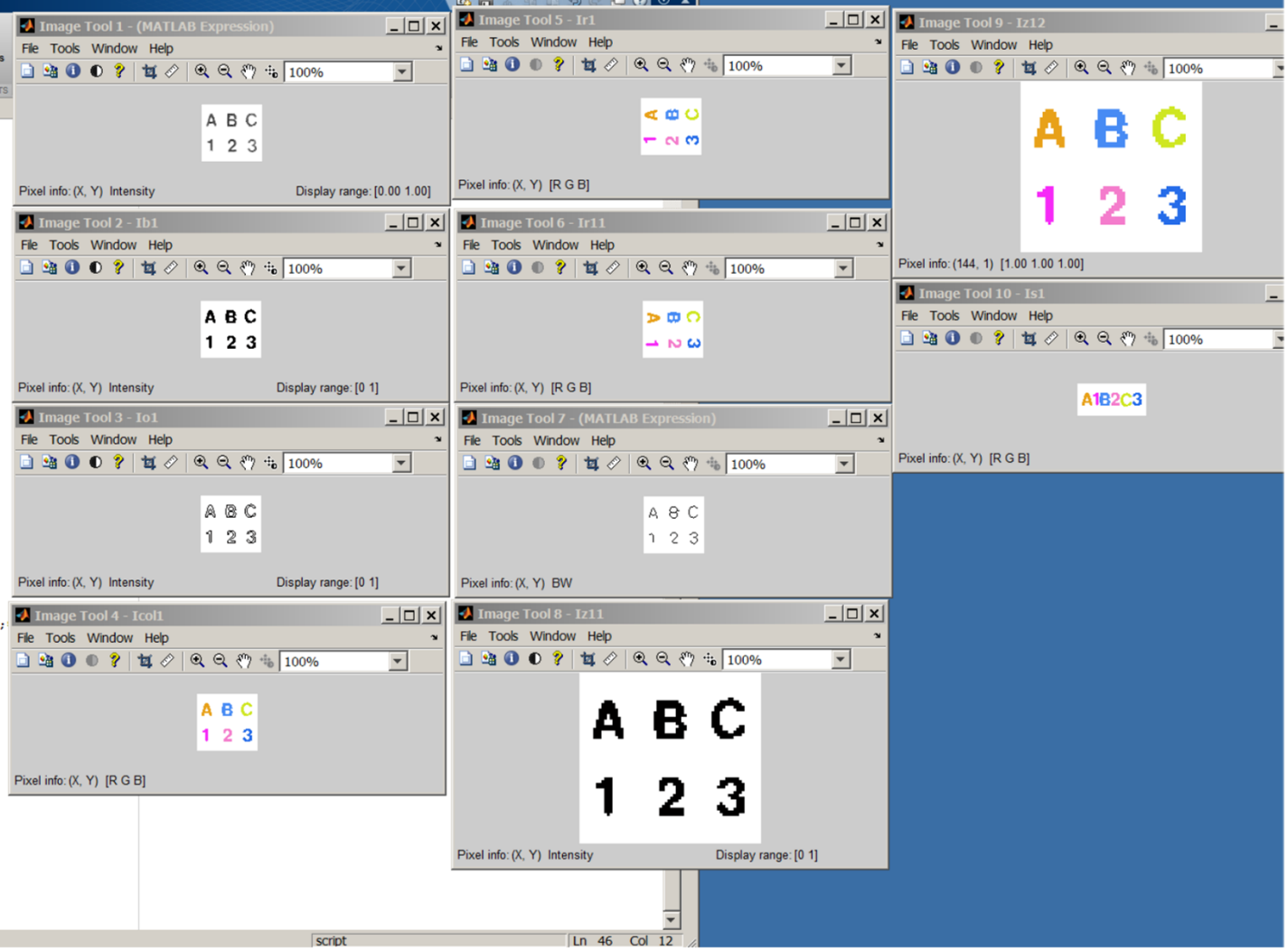
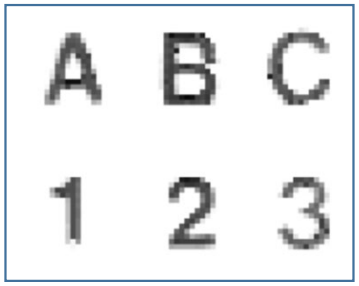
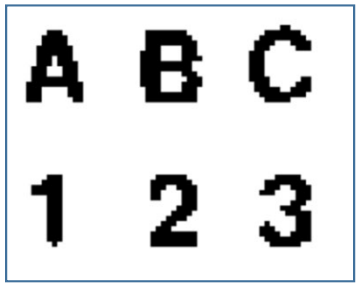


Image 2



Displaying image 1



Binary image

Rotation, 90 CW

Rotation, 90 CCW

Segmented image, each object is shown using a random colour

The outlines of the objects

Image with one-pixel-thick objects

Scaled image and translated objects

www.eng.nus.edu.sg/eitu/pc.html

What we are looking for in our grading:

Correctness of solution
Quality of solution
Effort
Proper report writing